

NYC 311 Operations Brief

NorthLedger Insights - automated analysis

NYC 311 OPERATIONS BRIEF - generated from table nyc_311 (22,542,090 rows, exact COUNT(*))

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1. SUMMARY

Metric	Value
Total rows	22,542,090
Distinct unique_key	22,542,090 (no duplicates)
Created date range	2020-01-01 through 2026-09-20 (data ex
Closed requests (closed_date present)	22,106,844 (98.07%)
Blank/null closed_date	435,246
Status mix	Closed 22,079,647; In Progress 283,218
Borough share	Brooklyn 30.00% (6,762,069); Queens 24

Annual volume: 2020: 2,942,024; 2021: 3,220,882; 2022: 3,169,960; 2023: 3,224,723; 2024: 3,456,769; 2025: 3,655,040; 2026 through Sep 20: 2,872,692.

Data cleaning notes: complaint type lives in column problem_formerly_complaint_type (bulk export rename). Dates stored 'MM/DD/YYYY HH:MI:SS AM/PM'; normalized to ISO via substr before julianday. All created_date values well-formed (length 22, none blank). 1,480,137 closed rows have closed_date earlier than created_date (clock skew/backfill); excluded from resolution stats. Medians and P90 computed by window row_number (nearest rank), never means.

2. TOP 10 COMPLAINT TYPES (normalized UPPER(TRIM); share of 22,542,090)

Rank	Complaint type	Count	Share %
1	ILLEGAL PARKING	2,927,763	12.99
2	NOISE - RESIDENTIAL	2,558,566	11.35
3	HEAT/HOT WATER	1,653,952	7.34
4	NOISE - STREET/SIDEWALK	1,167,728	5.18
5	BLOCKED DRIVEWAY	1,075,484	4.77
6	UNSANITARY CONDITION	707,282	3.14
7	REQUEST LARGE BULKY ITEM COLLECTION	432,048	2.80
8	STREET CONDITION	514,669	2.28
9	PLUMBING	428,196	1.90
10	WATER SYSTEM	422,863	1.88
-	Top 10 combined	14,089,441	62.51

3. AGENCY RESOLUTION TIME - 5 BUSIEST AGENCIES (closed requests, duration >= 0, median and P90 in days, window/nearest-rank)

Agency	Total requests	Closed rows used	Median days	P90 days	Note
NYPD	9,861,405	8,836,268	0.037 (about 53 minutes)	0.355 (about 18 hours)	Excludes 1,023,448 negative
HPD	4,455,776	4,264,724	4.725	41.781	Slowest tail; violation-issuance
DSNY	2,526,968	2,437,731	2.452	10.400	-
DOT	1,512,906	1,378,062	2.092	23.103	-

DEP	1,201,270	1,103,911	1.082	8.813	-
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4. MONTHLY COUNTS - LAST 48 COMPLETE MONTHS (2022-09 through 2026-08, exact COUNT by month)

Month	Count
2022-09	272,054
2022-10	267,072
2022-11	246,539
2022-12	260,662
2023-01	242,173
2023-02	225,002
2023-03	251,716
2023-04	247,496
2023-05	280,932
2023-06	267,168
2023-07	289,330
2023-08	282,757
2023-09	283,517
2023-10	302,769
2023-11	280,046
2023-12	271,817
2024-01	287,186
2024-02	240,436
2024-03	266,604
2024-04	267,565
2024-05	283,982
2024-06	306,200
2024-07	298,936
2024-08	284,823
2024-09	306,770
2024-10	306,803
2024-11	293,515
2024-12	313,949
2025-01	348,180
2025-02	255,364
2025-03	281,221
2025-04	272,549
2025-05	295,058
2025-06	306,442
2025-07	315,883
2025-08	304,039
2025-09	302,684
2025-10	336,612
2025-11	304,905

2025-12	332,103
2026-01	348,511
2026-02	334,690
2026-03	342,388
2026-04	302,191
2026-05	331,979
2026-06	334,833
2026-07	343,109
2026-08	328,828

Note: 2026-09 is partial at 206,163 through Sep 20 (data cut-off); excluded from model fit.

5. 12-MONTH FORECAST (OLS trend + monthly seasonality; 48-month history; MAPE fit 3.27%; residual sigma 14,806; 80% band)

Month	Forecast	80% low	80% high	Seasonal-naive
2026-09	346,583	327,608	365,558	302,684
2026-10	358,641	331,806	385,475	336,612
2026-11	336,578	303,712	369,444	304,905
2026-12	349,960	312,010	387,910	332,103
2027-01	361,839	319,410	404,269	348,511
2027-02	319,200	272,721	365,679	334,690
2027-03	340,809	290,606	391,012	342,388
2027-04	327,777	274,108	381,446	302,191
2027-05	353,315	296,390	410,240	331,979
2027-06	358,988	298,983	418,992	334,833
2027-07	367,141	304,208	430,074	343,109
2027-08	355,439	289,707	421,170	328,828
12-month total	4,176,270	approx 3,621,269	approx 4,731,269	3,942,833

Model vs baseline: model MAPE (fit) 3.27% versus seasonal-naive holdout MAPE 8.83% on the last 12 months. The model wins because it captures the +12.1% year-over-year growth (2026 Jan-Aug 2,666,529 vs 2025 Jan-Aug 2,378,736) that seasonal-naive structurally under-forecasts (naive missed Feb 2026 by 23.7% and Mar 2026 by 17.9%). Model total runs 5.9% above naive.

Honesty flag: partial Sept 2026 actuals run about 10,300/day (206,163 in 20 days), implying roughly 309,000-320,000 for the full month - below the model point of 346,583 and near/below the 80% low band. Early signal the model may run hot near-term; verify against October actuals before committing capacity plans.

6. LIMITATIONS

- 2026-09 is partial (through Sep 20); all figures for that month are incomplete.
- 1,480,137 closed rows (6.7% of closed) have closed_date earlier than created_date and were excluded from resolution stats; if these are clock-skew artifacts, medians are essentially unchanged, but the NYPD share affected (about 10.4% of its closed rows) suggests backfilling in NYPD data.
- Percentiles are nearest-rank on cleaned durations (≥ 0), computed per agency via window row_number; they are not means, so extreme outliers (e.g., DSNY max 3,930 days) do not distort the reported medians.
- Forecast residual sigma (14,806) gives an 80% band of roughly +/- 19,000 to 66,000 requests per month; the model assumes trend and seasonality continue and does not model policy changes, weather shocks, or reporting-channel shifts.
- Complaint type was normalized with UPPER(TRIM) to merge casing variants; legacy taxonomy labels (e.g.,

'Noise' vs 'Noise - Residential') belong to different eras and are counted separately.